Isogeny-class index-calculus yield: ECCp-131 and ECC2K-130

Characterization study report and visuals (branch claude/isogeny-class-ic-yield-20261102; generated by report/gen/gen_report.rs, interim until every cell has finished)

1. What this report is

Question (user, 2026-11-02): across the isogeny classes of ECC2K-130 and ECCp-131, do index-calculus relation yield, solve cost, resolving degree and related parameters differ between curves of one class? The study measures this with one instrument per part, the same on every curve, with an A/A noise floor, an isomorphic-model control, a negative-control population and, where the design has one, a planted positive control (DESIGN.md; PROTOCOL.md of each part).

- **Characterization study, not an improvement.** The preregistered expectation is a null. A null closes a lead. Nothing here is an improvement over the current index-calculus baseline, and the CLAUDE.md success criterion (every verified, measured improvement over that baseline) is not engaged: no baseline cost is changed, no phase is sped up.
- **No leaderboard row and no benchmarks row.** There is no cross-method claim and no matched rho reference, so the validation gate (replay certificate, validator report, matched rho with the 5a band) does not apply. `docs/ic-leaderboard.html`, `docs/ic/LEADERBOARD.md`, `docs/ic/leaderboard.json` and `benchmarks/README.md` are not touched. Per-row gaps to rho are not computed here, by design.
- **L0 isolation.** All three parts ran locally on a loaded host (P1 about 2 workers, P2 one ecbench cell at a time, P3 at most 2 workers, other agent sessions on the same 3 to 4 cores). Counted metrics (degrees, ranks, relation counts, field multiplications, row operations, solution counts) are exact and are the only ones that decide. Clocks (`cpu_s`, `wall_s`, `online_wall_ns_per_attempt`, `dreg_s`, `build_s`) are descriptive only and decide nothing.
- **Deferred, not dropped.** P1 tier 2 (`m2t4`), P2 at 36 and 40 bits (8 cells) and P3a-Y (n=43, k=13 yield line, claim C3) were not run under the amended protocols. They stay in the preregistration and are reported as "not run (deferred)", never filled.
- **Pending cells.** Cells still running at generation time show as "pending" in tables and graphs, never as nulls or zeros. Rerun `report/build.sh` when the runs finish.

2. Requirement to evidence

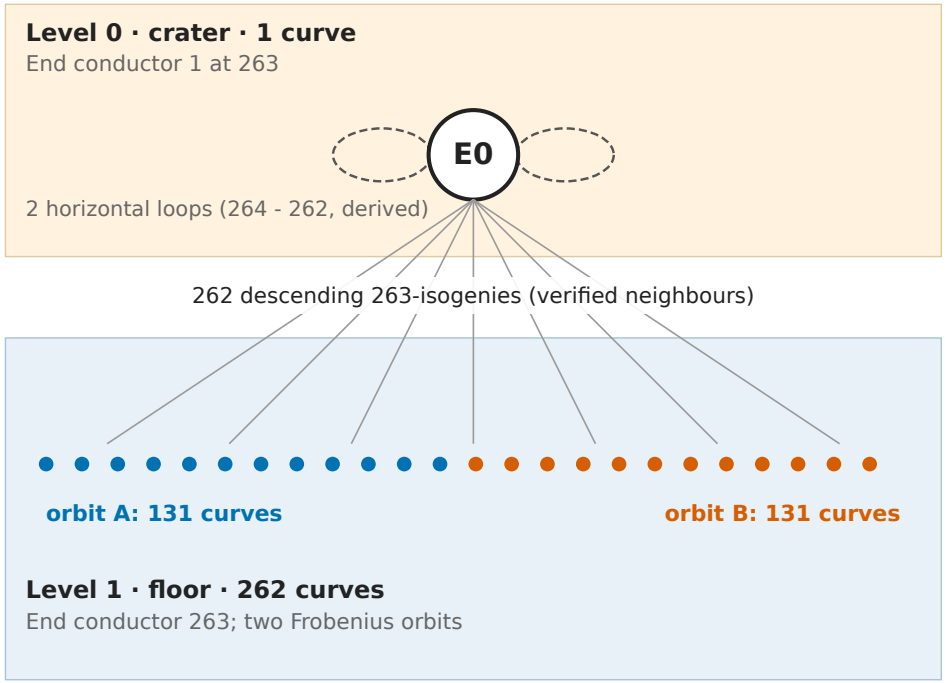
Requirement	Status	Evidence
Report states: characterization, not an improvement over the IC baseline	verified complete	section 1, DESIGN.md, three PROTOCOL.md
No leaderboard or benchmarks row; no rho comparison	verified complete	section 1; P2 PROTOCOL section 9, P3 PROTOCOL section 1
L0 isolation stated (counted exact, clock descriptive)	verified complete	section 1 and 3; P1 A1 section 14, P2 A1 and 10.1, P3 A1
Deferred cells stated (P1 tier 2, P2 36/40-bit, P3a-Y)	verified complete	sections 1 and 8
P1 ECCp-131 tier 1, arm-ratio strip plot	complete at tier-1 scope (tier 2 deferred)	section 5, figure 2, `p1_eccp131/runs/tier1_analysis.json`
P2 per-cell yield per curve, A/A band, planted control	partial: 10 of 12 run-scope cells finished	section 6, figure 3, `p2_prime_analogues/runs/<cell>/analysis.json`
P3 Dreg per curve and presentation family, E0 highlighted, standard V versus b-adapted	partial: 16 of 16 cards completed (receipt); more output present	section 7, figures 4 and 5, `p3_ecc2k130/full/`
Class-structure diagram (ECC2K-130 volcano levels beside ECCp-131 crater)	verified complete (structure from DESIGN.md)	section 3, figure 1
Generator, REPORT.md, REPORT.typ, REPORT.pdf, SVGs	see section 9	`report/`

3. Class structure and measurement conditions

Class structure: ECC2K-130 (conductor 263 x p') beside ECCp-131 (conductor 1)

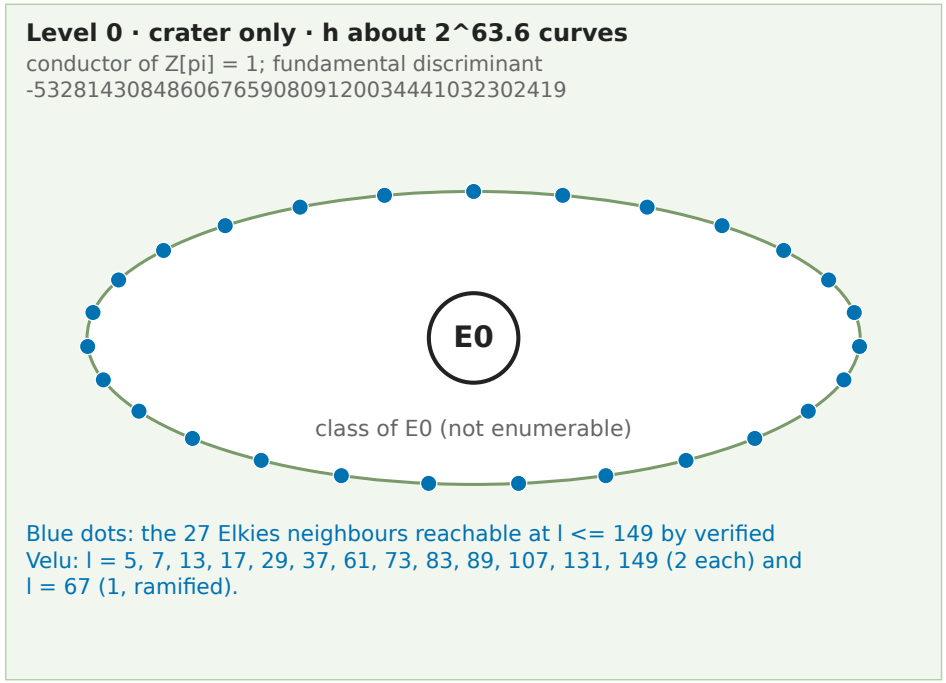
Facts from DESIGN.md (PARI 2.17, agrees with ecdb curve 22) and p3_ecc2k130/data/ecc2k130_curves.tsv. Drawn structure, not measurements. p' = 146505763881528721, about 2^57.0; only the 263 part of the conductor is walked.

ECC2K-130: 263-isogeny volcano, depth 1



E0: b = 1 in F_2 (Koblitz curve), trace -22283658519494248867, CM by Z[(1+sqrt(-7))/2]. Isogenies preserve the group order; the descended system depends on a curve through b only. Studied here (P3): Dreg, yield and F4 time on E0 and the 262 floor curves (n = 131), and the n = 43 analogue (258 level curves, 6 orbits).

ECCp-131: one crater, no vertical levels



Outside the class (null population, other group orders)

27 random non-isogenous curves; the isomorphic model of E0 is the same curve

Not a measurement figure: no data file. Facts in the tables of REPORT.md section 3.

Figure 1: Isogeny-class structure the instruments are applied to. Left: ECC2K-130 as a 263-isogeny volcano of depth 1 (E0 on the crater, 262 curves on the floor in two Frobenius orbits). Right: ECCp-131, a single crater of about 2^63.6 curves with 27 Elkies neighbours reachable at l <= 149, and the 27 random non-isogenous controls outside the class. Source: DESIGN.md, p3_ecc2k130/data/ecc2k130_curves.tsv.

	ECC2K-130	ECCp-131
trace	−2228365851949424886 7	29529377007934231835
fundamental discriminant	−7	−532814308486067659080912003444103230241 9
conductor of Z[pi]	263 x 146505763881528721	1 (crater only, h about 2^63.6)
computable neighbours	262 at level 263	Elkies l <= 150: 5,7,13,17,29,37,61,73,83,89,107,131,149 (2 each), 67 (1, ramified): 27
scaled analogue measured	n = 43 (level 257/258 curves, 6 orbits), full width n = 131 for Dreg	P2: primes of 24 to 32 bits (36, 40 deferred), conductor 1, full class enumerated

Table 2: Class facts (computed 2026-11-02, PARI 2.17 via cypari2; agrees with ecdb curve 22; copied from DESIGN.md).

Part / cell	Host and isolation as recorded	Checks as recorded
P1 tier 1	2 workers, nproc 4, model name : Intel(R) Xeon(R) Processor @ 2.80GHz	rows 14016; binary sha256 c230a30013d85bd0; done 14016 jobs in 6883.4s
P2 c24A_m2	ecbench isolation level recorded on rows: L1: 93, L2: 6435	audit: OK, 8160 records, 12 replays; in-session A/A: counts identical 2176/2176
P2 c24A_m3	ecbench isolation level recorded on rows: L1: 352, L2: 12704	audit: OK, 16320 records, 12 replays; in-session A/A: counts identical 4352/4352
P2 c24B_m2	ecbench isolation level recorded on rows: L1: 175, L2: 14609	audit: OK, 18480 records, 12 replays; in-session A/A: counts identical 4928/4928
P2 c24B_m3	ecbench isolation level recorded on rows: L1: 1761, L2: 27807	audit: OK, 36960 records, 12 replays; in-session A/A: counts identical 9856/9856
P2 c28A_m2	ecbench isolation level recorded on rows: L1: 1050, L2: 5478	audit: OK, 8160 records, 12 replays; in-session A/A: counts identical 2176/2176
P2 c28A_m3	ecbench isolation level recorded on rows: L1: 316, L2: 6212	audit: OK, 8160 records, 12 replays; in-session A/A: counts identical 2176/2176
P2 c28B_m3	ecbench isolation level recorded on rows: L1: 14784	audit: OK, 18480 records, 12 replays; in-session A/A: counts identical 4928/4928
P2 c32A_m2	ecbench isolation level recorded on rows: L1: 4352	audit: OK, 5440 records, 12 replays; in-session A/A: not run (dropped at 32 bits, PROTOCOL 4; round-parity A/A used)
P2 c32A_m3	ecbench isolation level recorded on rows: L1: 1002, L2: 3350	audit: OK, 5440 records, 12 replays; in-session A/A: not run (dropped at 32 bits, PROTOCOL 4; round-parity A/A used)
P2 c32B_m3	ecbench isolation level recorded on rows: L1: 3019, L2: 6837	audit: OK, 12320 records, 12 replays; in-session A/A: not run (dropped at 32 bits, PROTOCOL 4; round-parity A/A used)

Table 3: Measurement conditions read from the recorded files. The protocols accept L0 (loaded host); the ecbench level on P2 rows is what ecbench graded and is reported as recorded. In every case the clock is descriptive only.

4. Method in one paragraph per part

- **P1 (ECCp-131, full width, tier 1).** XL/Macaulay engine (`xl-incremental-sparse-echelon`) on the fixed bounded decomposition system, configs `m2t3` and `m3t2`, kinds null, planted and ladder. 58 arms: E0, an A/A repeat, an A/A permuted variable order-order arm, 1 isomorphic model, 27 Elkies neighbours, 27 random non-isogenous curves; 32 null targets, 8 planted targets, 3 reps, seed 20261102. Masked analysis frozen first, key opened after (A1 section 14).
- **P2 (prime analogues).** `ecbench` prime-abscissa mitm pipeline on every curve of an enumerated class (D = −84171, h = 128 for class A; D = −414651, h = 300 for class B), m = 2 and 3. Per curve pooled relation yield per attempt; ICC curve share of variance, max/min and max |z| against a 2000-permutation null; round-parity A/A halves; isomorphic-model control; planted +20% positive control (power must be at least 0.9 for a null to be informative).
- **P3 (ECC2K-130).** `p3h measure`: Weil-descended decomposition system over F_2, recorded `dreg` (resolving degree), exact `solutions`, rows and cols, for random V, standard V and b-adapted presentations, at n = 43 (all 258 level curves plus E0) and n = 131 (E0 plus 262 level-263 curves), k = 3 to 6.

5. P1: ECCp-131 at full width (tier 1)

`tier1_analysis.json` covers 14016 rows in 6 cells (config x kind), CI level 0.9995, 40000 bootstrap resamples. The analysis was produced with arm labels masked (`arm_labels_masked` = true); the key (sha256 b1463033d4d5bd5275e9b9b3202097fb34e72b9327e549f89413e0ec5766c96b) maps the masked labels to arms and is applied here only to name the points.

cell	arms vs E0	D_solve ratio	final_rank ratio	field_mults ratio	cpu_s ratio (descriptive)	AA_perm field_mults ratio	min identical-profile fraction
m2t3 ladder	1 (0 nb, 0 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	0.976 to 0.976	n/a	1.00
m2t3 null	57 (27 nb, 27 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	0.986 to 1.028	1.165	1.00
m2t3 planted	54 (25 nb, 26 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	0.974 to 1.074	1.124	1.00
m3t2 ladder	1 (0 nb, 0 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	1.042 to 1.042	n/a	1.00
m3t2 null	57 (27 nb, 27 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	0.968 to 0.995	0.976	1.00
m3t2 planted	18 (8 nb, 8 rnd)	1.000 to 1.000	1.000 to 1.000	1.000 to 1.000	0.973 to 1.015	0.976	1.00

Table 4: Measured (derived from the recorded per-arm geometric-mean ratios vs E0): range over arms other than AA_perm, per cell. AA_perm is the A/A arm with permuted variable order (re-encoding floor) and is listed separately.

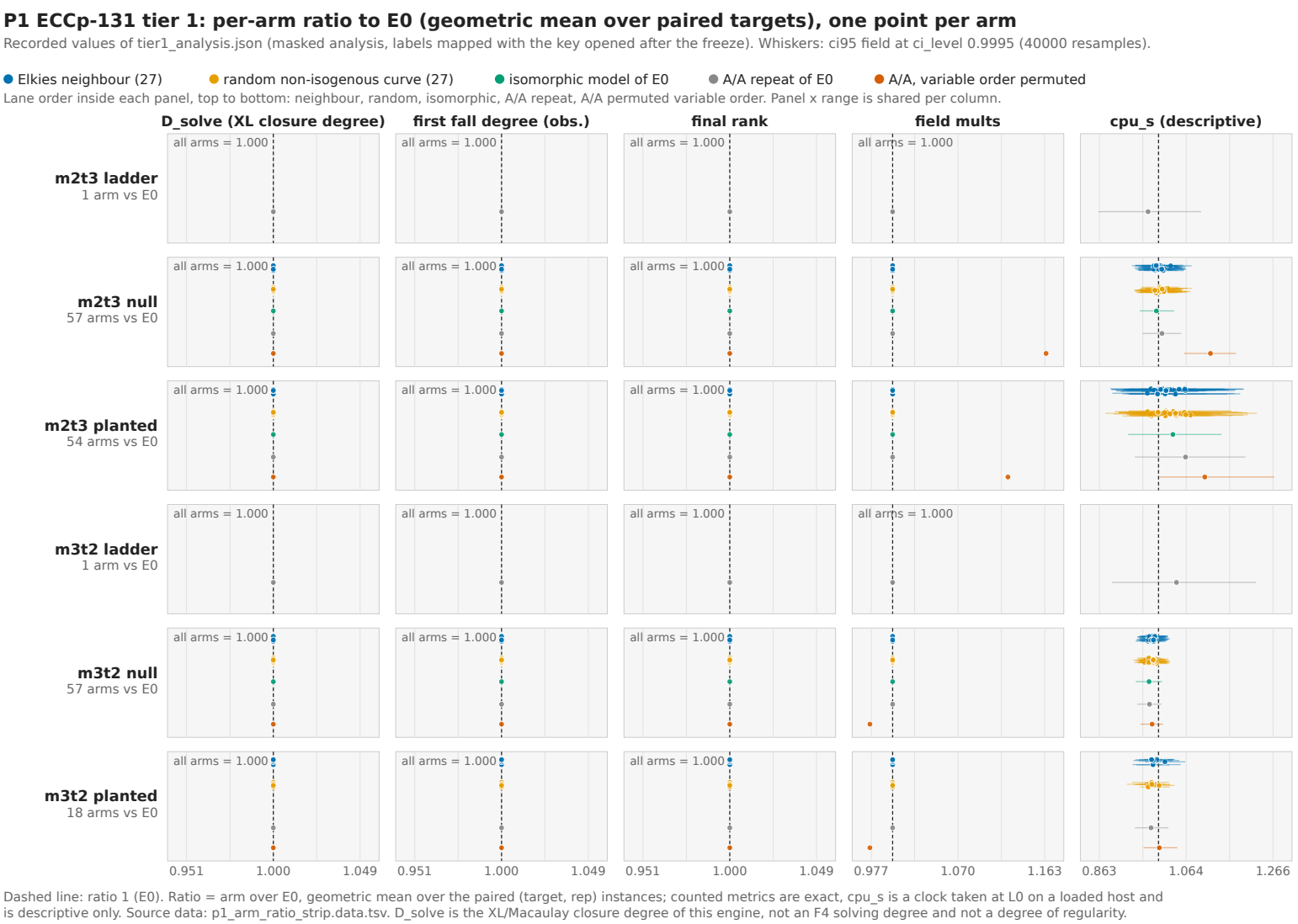


Figure 2: P1 arm-ratio strip plot. One point per arm and metric: ratio to E0 (geometric mean over paired target and rep instances), whiskers the recorded confidence interval. Counted metrics (D_solve, first fall degree, final rank, field mults) are exact; cpu_s is descriptive (L0, loaded host). Run: tier1 card CARD-isogeny-ic-yield-p1-tier1-20261102-1, analysis tier1_analysis.json.

config	kind	row status	rows
m2t3	ladder	solved	48
m2t3	null	solved	5568
m2t3	planted	not formed	72
m2t3	planted	solved	1320
m3t2	ladder	solved	48
m3t2	null	solved	5568
m3t2	planted	not formed	936
m3t2	planted	solved	456

Table 5: Recorded row status of tier1_rows.jsonl.gz (counted from the rows file).

Planted rows not formed: 1008. Recorded reason: planted instance not formed: factor base on this curve has fewer abscissae than m. Rows by config and role:

config	role	rows not formed
m2t3	neighbour	48
m2t3	random_curve_null_population	24
m3t2	iso_control	24
m3t2	neighbour	456
m3t2	random_curve_null_population	456

Table 6: These rows are missing observations (the factor base of that curve held fewer abscissae than m), not values; they are excluded from every ratio above.

cell	group	size	metrics with all values equal	metrics that vary (curve variance share)
m2t3 ladder	aa_E0_repeat_perm	2 curves, 8 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m2t3 null	aa_E0_repeat_perm	3 curves, 32 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.583), field_mults (share 1.000)
m2t3 null	class_E0_plus_neighbours	28 curves, 32 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m2t3 null	random_curves	27 curves, 32 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m2t3 planted	aa_E0_repeat_perm	3 curves, 8 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.199), field_mults (share 1.000)
m2t3 planted	class_E0_plus_neighbours	26 curves, 8 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000), field_mults (share 0.044)
m2t3 planted	random_curves	26 curves, 8 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.007), field_mults (share 0.017)
m3t2 ladder	aa_E0_repeat_perm	2 curves, 8 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m3t2 null	aa_E0_repeat_perm	3 curves, 32 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.067), field_mults (share 1.000)
m3t2 null	class_E0_plus_neighbours	28 curves, 32 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m3t2 null	random_curves	27 curves, 32 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.005)
m3t2 planted	aa_E0_repeat_perm	3 curves, 8 targets	D_solve, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000), field_mults (share 1.000)
m3t2 planted	class_E0_plus_neighbours	9 curves, 8 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)
m3t2 planted	random_curves	8 curves, 8 targets	D_solve, field_mults, final_cols, final_rank, final_rows, first_fall_degree_obs, n_standard_monomials_final	cpu_s (share 0.000)

Table 7: Recorded `icc` block of the analysis: for the A/A, class (E0 plus neighbours) and random-curve groups, which counted metrics take one value on every curve and, where they vary, the curve share of variance.

5.1 What it shows now

- **Measured.** In every cell, every arm other than AA_perm has ratio to E0 within 3.850e-5 of 1 on D_solve, first fall degree, final rank and field mults (the largest deviation over all cells and these four metrics is 3.850e-5; 0 means the recorded ratio is exactly 1). The arms present are the Elkies neighbours, the isomorphic model and the random non-isogenous curves (27 each, minus the planted instances that were not formed), for configs m2t3 and m3t2 and kinds null, planted and ladder.
- **Measured.** The arm that moves the operation count is AA_perm (field_mults ratio 0.976 to 1.165 over the cells in which it appears); its D_solve and final rank ratios deviate from 1 by at most 0. By construction (PROTOCOL section 3) this arm is the E0 instance with a seeded permutation of the variable order, the re-encoding noise floor: it sets how far the operation count moves under a re-encoding of the same curve, and it is not a curve effect.
- **Measured, descriptive.** `cpu_s` ratios of arms other than AA_perm lie within 0.074 of 1 (cells above); AA_perm cpu ratio 0.985 to 1.121. The clock was taken at L0 on a loaded host and decides nothing.
- **Measured.** 1008 planted rows were not formed (see table); they are missing, not zero.

Reading (main session, 2026-11-02; interpretation, labelled as such). At 131 bits the instrument's counted profile (XL closure degree, rank, field multiplications, per-degree profiles) is independent of the curve: it is the same on the 27 neighbours, on the isomorphic model and on the 27 random non-isogenous curves. Because random non-isogenous curves match as well, the null

population is uninformative about isogeny specifically: this instrument at this system size does not distinguish any curve from any other, so it cannot see an isogeny-class effect even if one existed in a quantity it does not count. `D_solve` is the XL/Macaulay closure degree of this study's engine; on the 147 comparable mirror systems of `p1_eccp131/CROSSCHECK.md` it is strictly above F4′s top step degree, so it is not F4′s solving degree and not a degree of regularity, and no statement here calls it one. Full-width yield is predicted, not observed (exact-curve screens found zero relations).

6. P2: prime analogues of ECCp-131′s class

class	m = 2	m = 3	class facts (from a finished cell)
c24A	finished (128 curves)	finished (128 curves)	p 2^23.70, D −84171, h 128
c24B	finished (300 curves)	finished (300 curves)	p 2^23.72, D −414651, h 300
c28A	finished (128 curves)	finished (128 curves)	p 2^27.31, D −84171, h 128
c28B	pending	finished (300 curves)	p 2^27.86, D −414651, h 300
c32A	finished (128 curves)	finished (128 curves)	p 2^31.94, D −84171, h 128
c32B	pending	finished (300 curves)	p 2^31.91, D −414651, h 300
c36A	not run (deferred)	not run (deferred)	-
c36B	not run (deferred)	not run (deferred)	-
c40A	not run (deferred)	not run (deferred)	-
c40B	not run (deferred)	not run (deferred)	-

Table 8: Cell inventory: 12 run-scope cells (24, 28, 32 bits, classes A and B, m = 2, 3) and 8 deferred cells (36 and 40 bits). Pending cells have no values; deferred cells were not run.

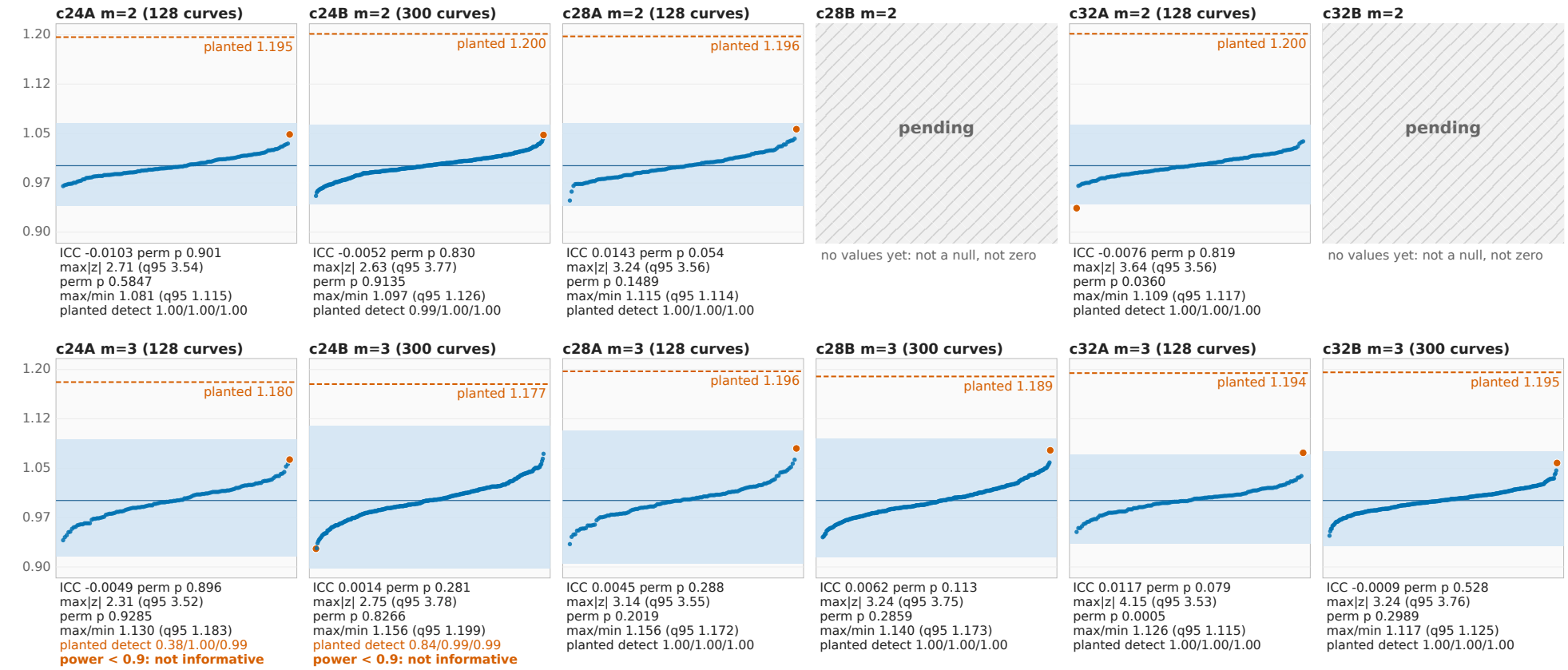


Figure 3: P2 yield per curve against the class mean with the A/A band and the planted-control effect, per finished cell (hatched panels are pending). Source: runs/<cell>/analysis.json (p2tool analyze, 2000 permutations, 2000 bootstrap resamples, seed 20261102).

cell	yield mean / attempt	ICC raw / boot upper95	ICC perm p	max abs z (null q95) p	max/min (null q95)	A/A sd / iso RMS (log ratio)	planted median pc/ main / min power	rel_gae ICC p	PROTOCOL 6 triggers (mechanical)
c24A_m2	0.0199	−0.0103 / 0.074	0.9005	2.71 (3.54) p 0.5847	1.081 (1.115)	0.0322 / 0.0336	1.195 / 1.00	0.471	none
c24A_m3	0.0330	−0.0049 / 0.038	0.8961	2.31 (3.52) p 0.9285	1.130 (1.183)	0.0484 / 0.0481	1.180 / 0.38	0.827	power < 0.9
c24B_m2	0.0198	−0.0052 / 0.073	0.8296	2.63 (3.77) p 0.9135	1.097 (1.126)	0.0316 / 0.0167	1.200 / 0.99	0.572	none
c24B_m3	0.0340	0.0014 / 0.041	0.2814	2.75 (3.78) p 0.8266	1.156 (1.199)	0.0528 / 0.0361	1.177 / 0.84	0.421	power < 0.9
c28A_m2	0.0199	0.0143 / 0.103	0.0540	3.24 (3.56) p 0.1489	1.115 (1.114)	0.0347 / 0.0202	1.196 / 1.00	0.450	none
c28A_m3	0.0329	0.0045 / 0.093	0.2884	3.14 (3.55) p 0.2019	1.156 (1.172)	0.0525 / 0.0430	1.196 / 1.00	0.090	none
c28B_m3	0.0330	0.0062 / 0.085	0.1134	3.24 (3.75) p 0.2859	1.140 (1.173)	0.0446 / 0.0445	1.189 / 1.00	0.193	none
c32A_m2	0.0010	−0.0076 / 0.080	0.8186	3.64 (3.56) p 0.0360	1.109 (1.117)	0.0326 / 0.0261	1.200 / 1.00	0.512	none
c32A_m3	0.0330	0.0117 / 0.102	0.0795	4.15 (3.53) p 0.0005	1.126 (1.115)	0.0370 / 0.0230	1.194 / 1.00	0.133	max|z| p < 0.0025
c32B_m3	0.0329	−0.0009 / 0.078	0.5282	3.24 (3.76) p 0.2989	1.117 (1.125)	0.0377 / 0.0278	1.195 / 1.00	0.449	none

Table 9: Recorded statistics of the frozen analysis for the finished cells. The last column only lists which numeric triggers of PROTOCOL section 6 are met (Bonferroni alpha 0.0025 over 20 tests; positive-control power below 0.9 means a null is not informative at 20%); the verdict per cell is assigned by the main session, and a trigger starts a follow-up protocol, it is not a claim.

Observation to carry into the reading. Cells whose last column lists a trigger are the ones the main session must read against the isomorphic-model control (presentation effect of the factor-base rule) before calling anything a curve effect. Cells that list `power < 0.9` are "not informative at 20%", never reported as nulls (PROTOCOL section 4).

7. P3: ECC2K-130 analogue and full width

card	what	output files (rows present)	state
a-ctrl-k3	P3a controls and C2 family, n=43, k=3	p3a_ctrl_k3: 3659 rows	completed (receipt)
a-main-k3	P3a main, n=43, k=3, 20 attempts	p3a_main_k3: 5180 rows	completed (receipt)
a-m2	P3a m=2, n=43, k=3,4,5,8	p3a_m2: 22000 rows	completed (receipt)
a-main-k5	P3a main, n=43, k=5 (bound)	p3a_main_k5: 518 rows	completed (receipt)
a-ctrl-k4-s0	P3a controls and C2 family, k=4, attempt 0	p3a_ctrl_k4.s0: 183 rows	completed (receipt)
a-ctrl-k4-s1	P3a controls and C2 family, k=4, attempt 1	p3a_ctrl_k4.s1: 183 rows	completed (receipt)
a-main-k4-s0	P3a main, k=4, attempt 0	p3a_main_k4.s0: 259 rows	completed (receipt)
a-main-k4-s1	P3a main, k=4, attempt 1	p3a_main_k4.s1: 259 rows	completed (receipt)
b-badapt	P3b b-adapted presentations, n=131, k=3..6	p3b_badapt: 2112 rows	completed (receipt)
b-bsparse	P3b b-sparse and nbadapt, n=131, k=5,6	p3b_bsparse: 200 rows	completed (receipt)
b-main-k5-s0	P3b random-V main, k=5, attempt 0	p3b_main_k5.s0: 263 rows	completed (receipt)
b-main-k5-s1	P3b random-V main, k=5, attempt 1	p3b_main_k5.s1: 263 rows	completed (receipt)
b-ctrl	P3b controls, CTRL131, k=3,4,5	p3b_ctrl: 216 rows	completed (receipt)
b-main-k34	P3b random-V main, k=3,4	p3b_main_k34: 1052 rows	completed (receipt)
b-stdv-s0	P3b standard V, curves 0..131 (E0 and orbit A)	p3b_stdv.s0a: 528 rows; p3b_stdv.s0b: 528 rows; p3b_stdv.s0c: 528 rows	completed (receipt)
b-stdv-s1	P3b standard V, curves 132..262 (orbit B)	p3b_stdv.s1a: 516 rows; p3b_stdv.s1b: 528 rows; p3b_stdv.s1c: 528 rows	completed (receipt)

Table 10: P3 card inventory. A file without a receipt may still be growing; its rows are included in the figures as present. P3a-Y (n=43, k=13 yield line, about 19.4 CPU-hours) is deferred, not run.

cell	isomorphic-model rows identical to main (dreg, solutions, refuted)	A/A rows identical to main	planted rows: planted_ok = 1 / 0 / not applicable (-1)
n=43, k=3	200/200	200/200	199 / 0 / 0
n=43, k=4	20/20	20/20	20 / 0 / 0
n=131, k=3	14/14	14/14	14 / 0 / 0
n=131, k=4	14/14	14/14	14 / 0 / 0
n=131, k=5	14/14	14/14	14 / 0 / 0

Table 11: PROTOCOL 6.1 instrument checks, recounted from the recorded rows (same (k, attempt, curve) key, same shard file). A cell with an identity below 100% is void by the protocol. Planted rows whose `outcome` is `atleast` are bounds and are not Dreg values.

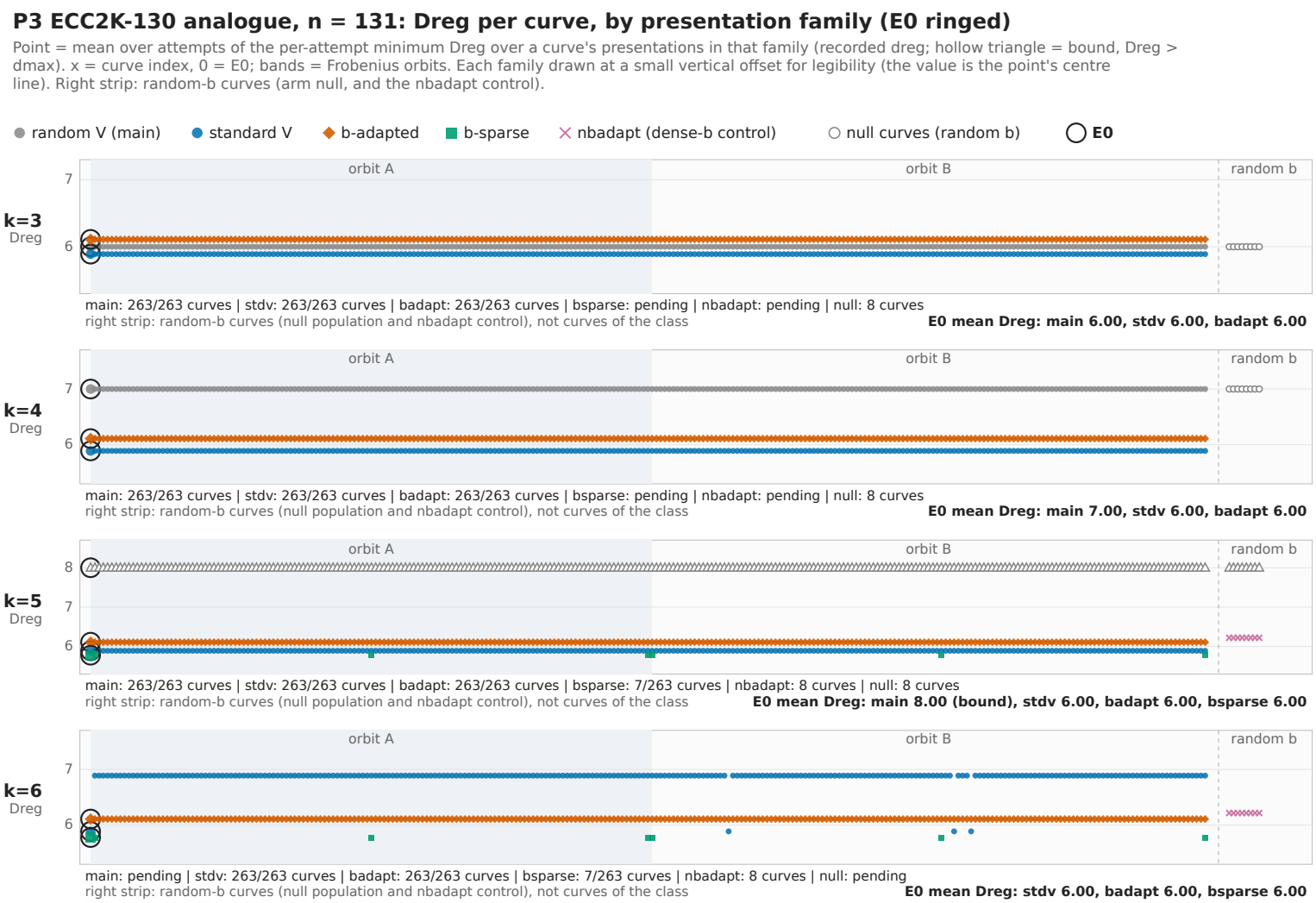


Figure 4: P3 at full width (n = 131): Dreg per curve for random V, standard V and b-adapted presentations, E0 ringed (curve 0), orbits A and B shaded. Pending series are named under each panel. Source: p3_dreg_n131.data.tsv from the recorded rows.

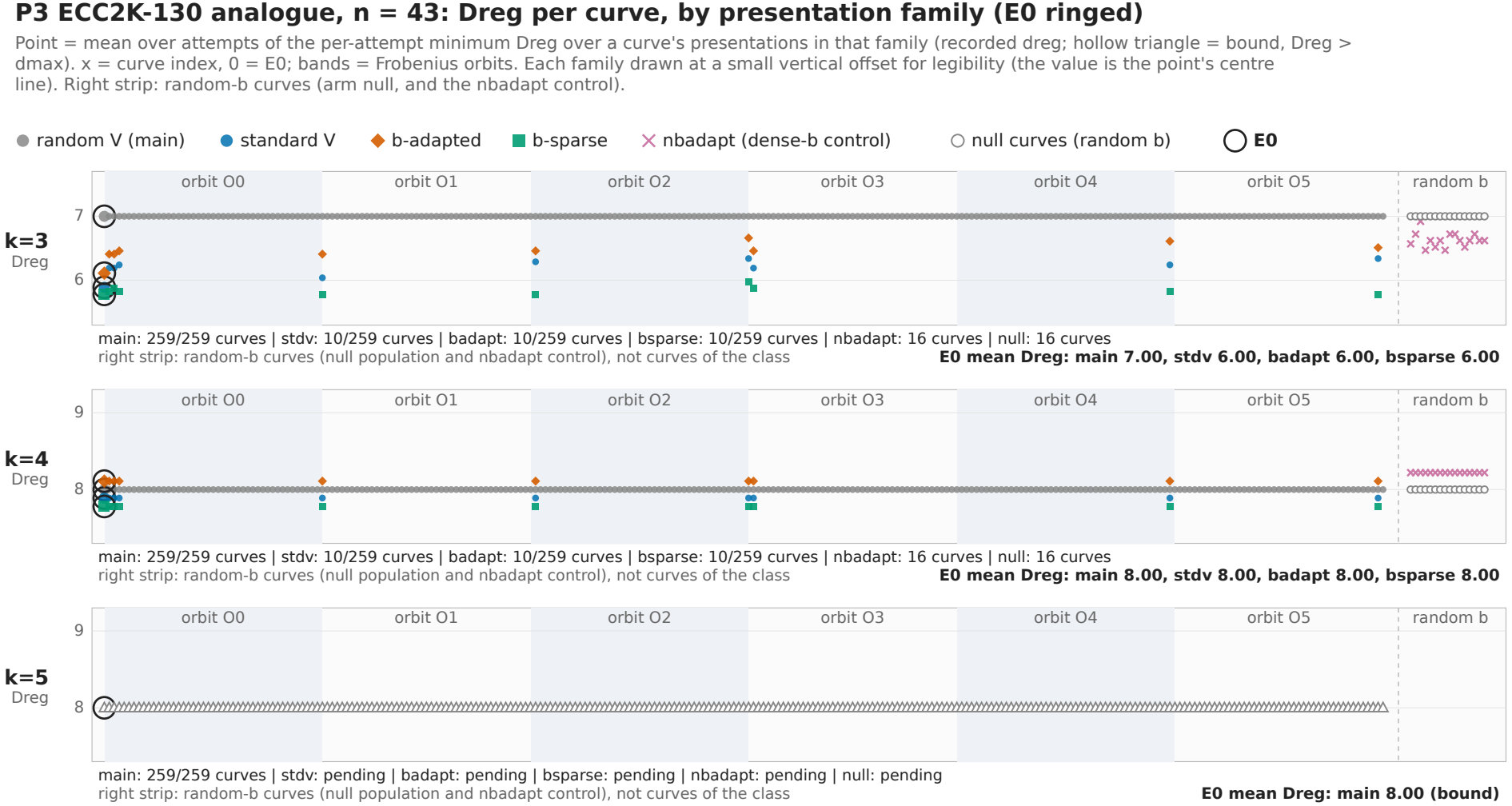


Figure 5: P3a analogue (n = 43, E0 plus 258 level curves, 6 orbits): Dreg per curve. The control and family arms were measured on a subset of curves (`--select`), the main arm on all curves. Hollow triangles are bounds (Dreg > dmax). Source: p3_dreg_n43.data.tsv.

cell	orbit	standard V only: floor-curve mean of min Dreg against E0	family minimum (arms present): floor-curve mean against E0	family arms with rows	family status
n=131, k=3	A	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=3	B	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=4	A	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=4	B	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=5	A	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=5	B	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=6	A	7.000 (range 7 to 7; 131 curve(s)); E0 6; difference 1.000	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=131, k=6	B	6.977 (range 6 to 7; 131 curve(s)); E0 6; difference 0.977	6.000 (range 6 to 6; 131 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O0	6.333 (range 6 to 7; 3 curve(s)); E0 6; difference 0.333	6.000 (range 6 to 6; 3 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O1	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O2	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O3	7.000 (range 7 to 7; 2 curve(s)); E0 6; difference 1.000	6.000 (range 6 to 6; 2 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O4	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=3	O5	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	6.000 (range 6 to 6; 1 curve(s)); E0 6; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=4	O0	8.000 (range 8 to 8; 3 curve(s)); E0 8; difference 0.000	8.000 (range 8 to 8; 3 curve(s)); E0 8; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=4	O1	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=4	O2	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=4	O3	8.000 (range 8 to 8; 2 curve(s)); E0 8; difference 0.000	8.000 (range 8 to 8; 2 curve(s)); E0 8; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)
n=43, k=4	O4	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	8.000 (range 8 to 8; 1 curve(s)); E0 8; difference 0.000	stdv, badapt, bsparse	all three family arms have rows (curve coverage in figures 4 and 5)

Table 12: Descriptive recorded-value summary, attempt 0 only, n = 43 family cells at k = 3 and 4 only (PROTOCOL A1) (the one attempt common to every family arm). Per curve: the minimum recorded Dreg over the presentations of the arms listed, per orbit: mean, range and count over curves, against E0 under the same procedure. This is not the frozen bootstrap analysis (`p3h analyze`, archived under report/data/p3_analysis/) and not PROTOCOL 6.3 outcome N or G: that reading needs the b-adapted and b-sparse arms to be complete, and cells marked incomplete must not be read as a family minimum gap.

The preregistered reading is C2: after minimising over the declared presentation family F on both sides, E0 and the floor curves coincide, with the single-presentation (standard V) gap present as the positive control (standard V, n = 131, k = 6: E0 Dreg 6 against floor Dreg 7). Where the table shows the single-presentation gap and an incomplete family, only the positive control is visible so far. A surviving gap would be read only as a statement about b in F_2 (E0 alone has b = 1), not about the isogeny class, and is not a speedup (no end-to-end S, no rho gap).

Main-session reading of the frozen pooled analyses (masked first, unmasked after the freeze; p3_ecc2k130/RESULTS.md): outcome N. At n = 131, k = 5 and 6 the presentation-family minimum is Dreg 6 for E0 and for all 262 level-263 curves on both attempts; the b-adapted presentation alone gives 6 on every curve, and random-b controls match. Random V: Dreg and matrix shape identical to E0 on every class curve at n = 131 (k = 3, 4); at n = 43 Dreg is identical and one target of 20 shows b-dependent column counts with E0 in the modal shape.

8. Deferred and pending

- **Not run (deferred), P1 tier 2:** config `m2t4`, 39 arms, 8 plus 2 targets (about 6.3 CPU-hours). In the preregistered design, no data taken under protocol version 1 amendment A1. Needs a further amendment before running.
- **Not run (deferred), P2 at 36 and 40 bits:** c36A_m2, c36A_m3, c36B_m2, c36B_m3, c40A_m2, c40A_m3, c40B_m2, c40B_m3 (8 cells). In the population and the preregistration; no data taken under amendment A1.
- **Not run (deferred), P3a-Y:** n = 43, k = 13 yield line (about 19.4 CPU-hours); claim C3 (yield against the Poisson prediction) is not run, and the `solutions` column of the cells that did run is descriptive.
- **Pending at generation time, P2:** c28B_m2, c32B_m2.
- **Pending or partial at generation time, P3:** none.
- Statements about pending cells are not made. In particular the C2 outcome (N or G) is open until the b-adapted and b-sparse arms and the remaining standard-V shards finish.

9. Reproduce, inputs and files

Rerun from the repository root: `research/isogeny_class_ic_yield_20261102/report/build.sh`. It compiles `report/gen/gen_report.rs` (Rust, std only, `rustc -O`), pools the P3 shards and archives the masked then unmasked `p3h analyze` outputs under `report/data/p3_analysis/` (needs the built `target/release/p3h` skipped with a note if absent), runs the generator and renders `REPORT.pdf` from `REPORT.typ` with Typst. Typst is document rendering only; if it is not installed the script installs the `typst` wheel in a scratch virtualenv outside the repository, as the research-visuals skill prescribes. No Python computes any value in this report.

- `report/REPORT.md`, `report/REPORT.typ`, `report/REPORT.pdf`: this report in three forms (generated, do not edit by hand).
- `report/figures/*.svg` with `*.data.tsv` beside each: class_structure, p1_arm_ratio_strip, p2_yield_per_curve, p3_dreg_n131, p3_dreg_n43.
- `report/data/inputs.sha256.tsv`: sha256 of every input file read in this build.
- `report/data/p3_analysis/`: `p3h analyze` text output per pooled P3 file family (masked first, then unmasked).

input file (relative to the study directory)	sha256 (first 16 hex; full in inputs.sha256.tsv)
p1_eccp131/runs/tier1.log.gz	1990ce69b4a4e4b6
p1_eccp131/runs/tier1_analysis.json	253419f2d7d0fe97
p1_eccp131/runs/tier1_analysis.json.key.json	b1463033d4d5bd52
p1_eccp131/runs/tier1_rows.jsonl.gz	a756d194ceb5998a
p2_prime_analogues/runs/c24A_m2/analysis.json	b196d2f04d4877e3
p2_prime_analogues/runs/c24A_m2/audit.json	0950bc0c45a18367
p2_prime_analogues/runs/c24A_m3/analysis.json	c4b4d91b3910b4ae
p2_prime_analogues/runs/c24A_m3/audit.json	5fde2045b1ee7f20
p2_prime_analogues/runs/c24B_m2/analysis.json	bd9cae34ea36de6e
p2_prime_analogues/runs/c24B_m2/audit.json	b13fa242a0337a28
p2_prime_analogues/runs/c24B_m3/analysis.json	40a5e5944275d87c
p2_prime_analogues/runs/c24B_m3/audit.json	28c25dc4762f0aaa
p2_prime_analogues/runs/c28A_m2/analysis.json	01e919a90cfd2725
p2_prime_analogues/runs/c28A_m2/audit.json	b7b9447ece6d7693
p2_prime_analogues/runs/c28A_m3/analysis.json	017d0ca14e8795c9
p2_prime_analogues/runs/c28A_m3/audit.json	002d0164ac54045d
p2_prime_analogues/runs/c28B_m3/analysis.json	1765c53edc0276c3
p2_prime_analogues/runs/c28B_m3/audit.json	f27c9e54faf6b29b
p2_prime_analogues/runs/c32A_m2/analysis.json	dba868b529f5322f
p2_prime_analogues/runs/c32A_m2/audit.json	af60cc5c2c3c00c2
p2_prime_analogues/runs/c32A_m3/analysis.json	b4e907dd0328bc41
p2_prime_analogues/runs/c32A_m3/audit.json	9906464becf9e795
p2_prime_analogues/runs/c32B_m3/analysis.json	2330be8d60844005
p2_prime_analogues/runs/c32B_m3/audit.json	3d394c5f8f2912ab
p3_ecc2k130/data/ecc2k130_curves.tsv	ee987b52191581be
p3_ecc2k130/full/p3a_ctrl_k3.jsonl	f136198871fd2e0f
p3_ecc2k130/full/p3a_ctrl_k4.s0.jsonl	38b5aa5f790b7d98
p3_ecc2k130/full/p3a_ctrl_k4.s1.jsonl	df920701d2ff3636
p3_ecc2k130/full/p3a_m2.jsonl	d4aba33321e915e2
p3_ecc2k130/full/p3a_main_k3.jsonl	5a74f7c788b3afb7
p3_ecc2k130/full/p3a_main_k4.s0.jsonl	ed54f2385dd64492
p3_ecc2k130/full/p3a_main_k4.s1.jsonl	483f319d6657d46e
p3_ecc2k130/full/p3a_main_k5.jsonl	e231989e04918260
p3_ecc2k130/full/p3b_badapt.jsonl	e4f3f7a44ab6293f
p3_ecc2k130/full/p3b_bsparse.jsonl	4529006fc66b534e
p3_ecc2k130/full/p3b_ctrl.jsonl	db555205a315ce34
p3_ecc2k130/full/p3b_main_k34.jsonl	027f656372a38aec
p3_ecc2k130/full/p3b_main_k5.s0.jsonl	0d349d7892f14204
p3_ecc2k130/full/p3b_main_k5.s1.jsonl	7d242d7bb016e554
p3_ecc2k130/full/p3b_stdv.s0a.jsonl	12726d01e4e2b785
p3_ecc2k130/full/p3b_stdv.s0b.jsonl	685559ff5d06aadc
p3_ecc2k130/full/p3b_stdv.s0c.jsonl	38501eaddf81c628
p3_ecc2k130/full/p3b_stdv.s1a.jsonl	51afd98ab060ba64
p3_ecc2k130/full/p3b_stdv.s1b.jsonl	48a22d0c68af26e4
p3_ecc2k130/full/p3b_stdv.s1c.jsonl	eb937a8bd2bcb6c2

Table 13: Inputs read by this build.